2.17 Tool holding Devices

Different tools are used for performing different operations. They are fitted into the drill spindle by different methods. They are

1. directly fitting in the spindle
2. a sleeve
3. a socket
4. a drill chuck
5. special attachments
6. Tapping attachments
7. Floating holder.

Directly fitting in the spindle

The following figure shows the drill directly fit into the spindle.

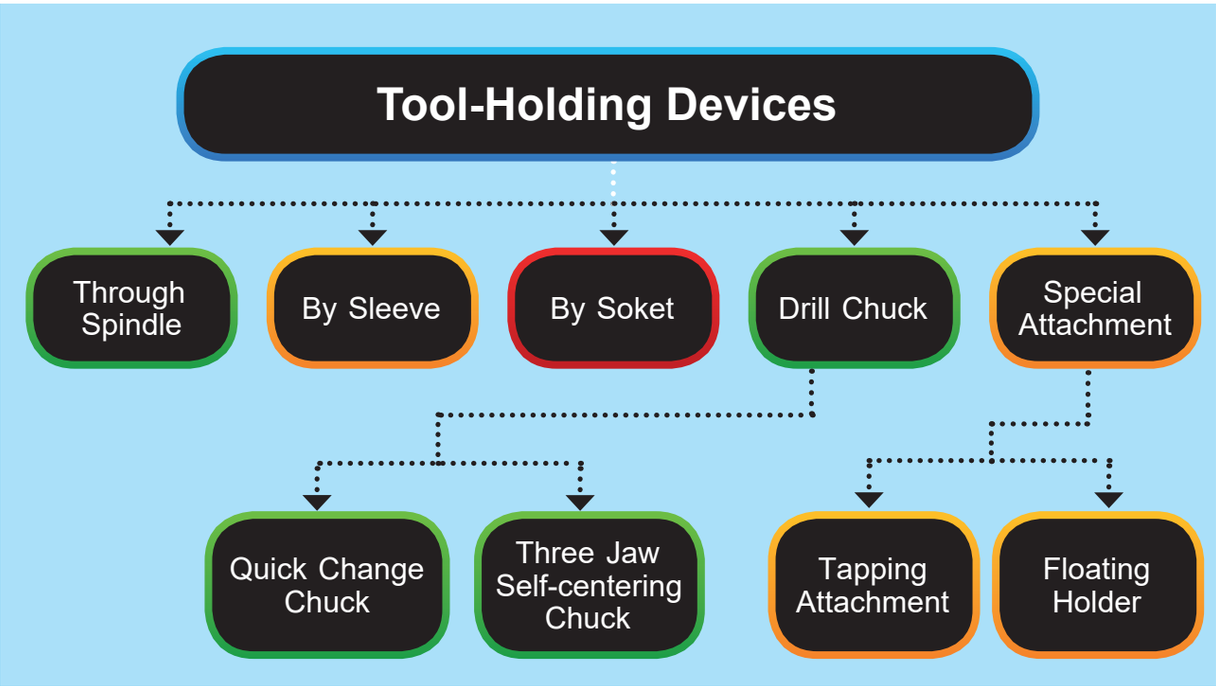
Almost all drilling machines have their spindle bored out to a standard taper (1:20) to receive the taper shank of the tool. While fitting the tool, the shank of the drill (or) any other tool is forced into the tapered hole and the tool is gripped by friction. The tool may be rotated with the spindle by friction between the tapered surface and the spindle. But to ensure a positive drive the tang of the tool fits into a slot at the end of the taper hole.

The tool may be removed by pressing a tapered wedge known as drift or key into the slotted hole of the spindle.

The drift is made up of hard steel.

Sleeve

The drill spindle is suitable for holding only one size of tool shank. If the shank of the tool is smaller than the taper in the spindle hole a taper sleeve is used. The outside taper of the sleeve confirms to the spindle taper and the inside taper holds the shanks of the smaller size tools. The sleeve has a flattened end or tang which fits into the slot of the spindle. The tang of the tool fits into a slot provided at the end of the taper hole of the sleeve. Different sizes of tool shanks may be held by using different sizes of sleeve. In order to remove the drill from the



XII BME_English version CHAPTER 02.indd 48 1/8/2020 1:44:54 PM